

Math 220: Homework 3

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Problem 1 (Examples and nonexamples of bases). (i) Prove that $\mathcal{B} = \{(p, q) : p < q \in \mathbb{Q}\}$ is a basis for the Euclidean topology on $\mathbb{R}$.

Proof. Let $(a, b) \in \mathcal{T}_E$ and let $x \in (a, b)$. Since $\mathbb{Q}$ is dense in $\mathbb{R}$, there exists p and q in $\mathbb{Q}$ such that $a < p < x < q < b$, implying that $x \in (p, q) \subseteq (a, b)$. Since $(p, q) \in \mathcal{B}$ and (a, b) is an arbitrary element of $\mathcal{T}_E$, it follows that $\mathcal{B}$ is a basis for the Euclidean Topology $\mathcal{T}_E$. ■

(ii) Prove that $\mathcal{B}' = \{[p, q) : p < q \in \mathbb{Q}\}$ is not a basis for the lower limit topology on $\mathbb{R}$.

Proof. Suppose for contradiction that $\mathcal{B}'$ is a basis for the lower limit topology. Define $a = \sqrt{3}$ and $b = \sqrt{3} + 1$. Then we see that $[a, b)$ is an element of the lower limit topology. Since $\mathcal{B}'$ is a basis for $\mathcal{T}_L$, there exists $B' \in \mathcal{B}'$ such that $a \in B' \subseteq [a, b)$ where $B' = [p, q)$ for some $p, q \in \mathbb{Q}$. Since $a \in [p, q)$, it follows that $p \leq a$. Since $[p, q) \subseteq [a, b)$, it follows that $a \leq p$ which tells us that $p = a$. But now p is an irrational number which is a contradiction. Hence, $\mathcal{B}'$ cannot be a basis for the lower limit topology. ■

(iii) Is $\mathcal{B}'$ from the previous point the basis for a topology on $\mathbb{R}$? Prove your claim.

Proof. We claim that $\mathcal{B}'$ from the previous part is the basis for the topology on $\mathbb{R}$. That is, we want to show that

(I) $\bigcup_{B \in \mathcal{B}'} B = \mathbb{R}$

(II) For all $A, B \in \mathcal{B}$, $A \cap B$ can be written as a union of elements of $\mathcal{B}'$.

Starting with (I), we see that it suffices to show that $\mathbb{R} \subseteq \bigcup_{B \in \mathcal{B}'} B$. To this end, let $x \in \mathbb{R}$. Since $\mathbb{Q}$ is dense in $\mathbb{R}$, there exists $p, q \in \mathbb{Q}$ such that $p < x < q$. That is, $x \in (p, q) \subseteq [p, q)$. Since $[p, q) \subseteq \bigcup_{B \in \mathcal{B}'} B$, it follows that $x \in \bigcup_{B \in \mathcal{B}'} B$. Thus, it follows that $\mathbb{R}$ is a subset of the union of elements of $\mathcal{B}'$. That is,

$$\bigcup_{B \in \mathcal{B}'} B = \mathbb{R}.$$

(II) Let $A, B \in \mathcal{B}$ be arbitrary. Our goal is to show that $A \cap B \in \mathcal{B}$. Since $A, B \in \mathcal{B}$, it follows that $A = [p_1, q_1)$ and $B = [p_2, q_2)$ where $p_1, p_2, q_1, q_2 \in \mathbb{Q}$. Let $x \in A \cap B$. Set $m = \max\{p_1, p_2\}$ and $\ell = \min\{q_1, q_2\}$. Then we get that $x \in [m, \ell) \subseteq A \cap B$. Since m and ℓ are in $\mathbb{Q}$, it follows that $[m, \ell)$ is an element of $\mathcal{B}'$. Implying that $A \cap B$ can be written in terms of a union of elements from $\mathcal{B}'$ and we are done.

Thus, (I) and (II) imply that $\mathcal{B}'$ is a basis for a topology on $\mathbb{R}$. ■

(iv) Is the family $\mathcal{B}'' = \{[a, b] : a, b \in \mathbb{R}, a < b\}$ the basis for a topology on $\mathbb{R}$? Prove your claim.

Proof. The family above is not a basis for a topology on $\mathbb{R}$. Suppose for a contradiction that it is. Then $A = [0, 1]$ and $B = [1, 2]$. Then $A \cap B = \{1\}$. Since $\mathcal{B}''$ is a basis for a topology on $\mathbb{R}$, then there exists $[c, d] \in \mathcal{B}''$ such that $1 \in [c, d] \subseteq A \cap B = \{1\}$. But this would imply that $\{1\}$ contains more than one point which is absurd. Hence, $\mathcal{B}''$ cannot be a basis for a topology on $\mathbb{R}$. ■

Problem 2 (The K -topology on $\mathbb{R}$). Let

$$\mathcal{B}_K = \{(a, b) : a, b \in \mathbb{R}, a < b\} \cup \{(a, b) \setminus K : a, b \in \mathbb{R}, a < b\}$$

where $K = \{1/n : n \in \mathbb{N}\}$.

- (i) Prove that $\mathcal{B}_K$ is the basis for a topology on $\mathbb{R}$. We will call this topology the K -topology and denote it by $\mathcal{T}_K$.

Proof. We will show that

$$(I) \bigcup_{B \in \mathcal{B}} B = \mathbb{R}$$

(II) For all $A, B \in \mathcal{B}_K$, for all $x \in A \cap B$, there exists a $C \in \mathcal{B}_K$ such that $x \in C \subseteq A \cap B$.

Starting with (I), let $x \in \mathbb{R}$. Then

$$x \in (x-1, x+1) \subseteq \bigcup_{x \in \mathbb{R}} (x-1, x+1) = \mathbb{R}$$

where $(x-1, x+1)$ is in $\mathcal{B}_K$. Hence, $\bigcup_{B \in \mathcal{B}_K} B = \mathbb{R}$.

Next, we will show that $\mathcal{B}_K$ satisfies (II). Let $A, B \in \mathcal{B}_K$ be arbitrary and let $x \in A \cap B$. We will consider three cases; that is,

- (1) $A, B \in \{(a, b) : a, b \in \mathbb{R}, a < b\}$
- (2) $A \in \{(a, b) : a, b \in \mathbb{R}, a < b\}$ and $B \in \{(a, b) \setminus K : a, b \in \mathbb{R}, a < b\}$
- (3) $A, B \in \{(a, b) \setminus K : a, b \in \mathbb{R}, a < b\}$.

(1) Suppose $A, B \in \{(a, b) : a, b \in \mathbb{R}, a < b\}$, then we get that $A = (a_1, b_1)$ and $B = (a_2, b_2)$ for some $a_1, a_2, b_1, b_2 \in \mathbb{R}$. Let $m = \max\{a_1, a_2\}$ and $\ell = \min\{b_1, b_2\}$. Let $x \in A \cap B$. Define $C = (m, \ell)$. Then it follows that $x \in C = (m, \ell) \subseteq A \cap B$. Hence, we see that $A \cap B$ can be written as a union of elements in $\mathcal{B}_K$.

Suppose that (2) holds. Then $A = (a_1, b_1)$ and $B = (a_2, b_2) \setminus K$. Note that

$$A \cap B = A \cap (B \setminus K) = (A \cap B) \setminus K.$$

Using the same m and ℓ from (1), it follows that $(a_1, b_1) \cap (a_2, b_2) = (m, \ell)$. Choosing $\varepsilon = \frac{1}{2} \min\{x-m, \ell-x\}$, it follows that $(x-\varepsilon, x+\varepsilon) \subseteq (m, \ell)$. Define $C = (x-\varepsilon, x+\varepsilon) \setminus K$ which is in $\mathcal{B}_K$. Then $x \in C \subseteq (m, \ell) \subseteq A \cap B$ and we are done.

(3) Suppose $A, B \in \{(a, b) \setminus K : a, b \in \mathbb{R}, a < b\}$. Then $A = (a_1, b_1) \setminus K$ and $B = (a_2, b_2) \setminus K$. Then we have

$$\begin{aligned} A \cap B &= \left((a_1, b_1) \setminus K \right) \cap \left((a_2, b_2) \setminus K \right) \\ &= \left[(a_1, b_1) \cap (a_2, b_2) \right] \setminus K \\ &= (m, \ell) \setminus K. \end{aligned}$$

Using the same choice of ε as in the previous case, we can define $C = (x-\varepsilon, x+\varepsilon) \setminus K$ which is a subset of $A \cap B$. Hence, $x \in C \subseteq A \cap B$ and we are done. Now, (I) and (II) imply that $\mathcal{B}_K$ is a basis for some topology on $\mathbb{R}$. ■

- (ii) Prove that $\mathcal{T}_K$ is strictly finer than the Euclidean Topology.

Proof. Note that $(a, b) \in \mathcal{T}_E$ is also in $\mathcal{T}_K$ since (a, b) is in $\mathcal{B}_K$ (basis for $\mathcal{T}_K$). However, $(0, 1) \setminus K \notin \mathcal{T}_E$ is not open with respect to $\mathcal{T}_E$ since every neighborhood of $1/n$ contains points of K ; that is, $(0, 1) \setminus K$ fails to contain a Euclidean interval around any $1/n$. Hence, we see that $\mathcal{T}_K$ is strictly finer than the Euclidean Topology. ■

- (iii) Prove that $\mathcal{T}_K$ and the lower limit topology are not comparable.

Proof. Our goal is to find an open set in $\mathcal{T}_L$ that is not in $\mathcal{T}_K$ and an open set in $\mathcal{T}_K$ that is not in $\mathcal{T}_L$.

Notice that $(0, 1) \setminus K$ is open with respect to the K -topology but not open in the lower limit topology because any neighborhood $[x, x + r)$ (for any $r > 0$) around 0 will contain points of the form $1/n$. Hence, $(0, 1) \setminus K$ cannot be in $\mathcal{T}_L$.

Lastly, we have $[0, 1)$ is open with respect to the lower limit topology $\mathcal{T}_L$. Consider $x = 0$ and notice that every neighborhood around 0 in the K -topology must have points that are less than 0. That is, every neighborhood with (neighborhoods in the form (a, b) or $(a, b) \setminus K$) with respect to the K -topology that contain 0 will not be completely contained in $[0, 1)$. Thus, $[0, 1)$ is not in $\mathcal{T}_K$. Thus, we conclude that the K -topology and the lower limit topology are not comparable. ■

Problem 3 (A set is open if and only if it is a neighborhood of all its points). Let X be a topological space and $A \subseteq X$. Prove that A is open if and only if $A \in \mathcal{N}(x)$ for all $x \in A$.

Proof. ($\implies$) Let $x \in A$ be arbitrary. Since A is open, set $U = A$ such that $x \in A \subseteq U$ (since A is contained within itself). By definition, this tells us that $A \in \mathcal{N}(x)$.

($\impliedby$) Suppose $A \in \mathcal{N}(x)$ for all $x \in A$. Our goal is to show that A is an open set. By assumption, for all $x \in A$, there exists U_x such that $x \in U_x \subseteq A$. Since $U_x \subseteq A$ for all $x \in A$, it follows that

$$\bigcup_{x \in A} U_x \subseteq A.$$

But we also have that $A \subseteq \bigcup_{x \in A} U_x$ (since for all $x \in A$, $x \in U_x$ and U_x is a subset of the union $\bigcup_{x \in A} U_x$) and so we conclude that

$$\bigcup_{x \in A} U_x = A.$$

Since U_x is open for all $x \in A$ and the arbitrary union of open sets is open, it follows that A must also be open. ■

Problem 4 (Examples of interiors of sets). (i) It can be shown that $\mathcal{B}_R = \{(a, b) : a, b \in \mathbb{R}, a < b\}$ is the basis for a topology on $\mathbb{R}$ called the **upper limit topology**. In this topology, find the interior of the sets $A = (a, b)$ and $B = [a, b]$ where $a, b \in \mathbb{R}, a < b$. Prove the claims.

Proof. We claim that $\overset{\circ}{A} = A$ and $\overset{\circ}{B} = (a, b]$.

Starting with the first claim, we know that A is open with respect to the Euclidean topology. That is, for any $x \in A$, we can find a $\delta > 0$ such that $(x - \delta, x + \delta) \subseteq (a, b)$. But note that $(x - \delta, x + \delta) \subseteq (a, b) = A$ and so every point in (a, b) is an interior point with respect to the upper limit topology. Hence, $\overset{\circ}{A} = A$.

With the second claim, we notice that $(a, b] \subseteq [a, b]$ is the largest open set in the upper limit topology that is contained in $[a, b]$ since if we add a into the set, then $[a, b]$ will no longer be open with respect to the upper limit topology (that is, any neighborhood around a will not be contained in $[a, b]$). Hence, it follows that $\overset{\circ}{B} = (a, b]$. ■

(ii) In $\mathbb{R}$ equipped with the cofinite topology, find the interior of the following sets: $A = (a, b)$, $B = (a, \infty)$, $C = (-\infty, a) \cup (a, \infty)$, where $a, b \in \mathbb{R}, a < b$. Prove your claims.

Proof. Recall that in the cofinite topology, any set whose complement is finite must be an open set. We claim that $\overset{\circ}{A} = \emptyset$, $\overset{\circ}{B} = \emptyset$ and $\overset{\circ}{C} = C$.

Starting with our first claim. We see that $A^c = (-\infty, a] \cup [b, +\infty)$ is an infinite set because $(-\infty, a]$ and $[b, +\infty)$ are infinite sets. Hence, the largest open set that is contained in A must be the empty set $\emptyset$. Hence, $\overset{\circ}{A} = \emptyset$.

Next, we see that $B^c = [(a, +\infty)]^c = (-\infty, a]$ is also an infinite set. Hence, $\overset{\circ}{B} = \emptyset$. Finally, we see that C^c is a finite set because

$$\begin{aligned} C^c &= [(-\infty, a) \cup (a, +\infty)]^c \\ &= (-\infty, a)^c \cap (a, +\infty)^c \\ &= [a, +\infty] \cap (-\infty, a] \\ &= \{a\}. \end{aligned}$$

Hence, it follows that $C \in \mathcal{T}$ and so $C = \overset{\circ}{C}$. ■